



Department of Computer Science and Engineering

Academic Year: 2023- 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3591 Computer Networks

Name of the Faculty member (s): Mrs.M.Swarna Sudha

Innovative Practice Description

Unit / Topic: Unit I / Domain Name System

Course Outcome: CO 1

Activity Chosen: Collaborative Learning

Date of Implementation: 21.08.2023

Justification:

Collaborative learning is essential in the context of DNS (Domain Name System) in networking as it addresses the workings of internetworking. DNS involves complex hierarchies and configurations, and collaborative learning allows students to collectively navigate these challenges, pooling their understanding and sharing diverse perspectives. Moreover, the diverse skills and knowledge within a collaborative setting provide a comprehensive understanding of DNS, while the interactive nature of such activities increases engagement and motivates students to actively participate in the learning process. Collaborative learning in DNS not only promotes teamwork but also enhances communication skills and establishes a supportive learning environment, essential elements for success in the dynamic field of networking.

Time Allotted for the Activity: 30 minutes

Details of the Implementation

- The instructor took twenty minutes to thoroughly clarify each key concept in the classroom. Conduct an introductory session to familiarize the students with DNS concepts, its importance in DNS, and its role in translating domain names to IP addresses.
- After the initial introduction, students like Anitharani K and Birunthalakshmi B actively participated in a discussion, asking questions and getting clarification on the working of DNS.
- Students were asked to form groups based on their interests, and they chose to create 9 teams, with 5 teams consisting of girls and 4 teams of boys.

- Each group was given a specific DNS-related topic based on what they liked and were good at, covering areas like DNS basics, DNS Architecture, record types, security, and DNS Troubleshooting.
- Devishenbaga R and team focused on DNS Basics and Architecture, while D Kanishkashree and Gokila G worked on DNS Record Types. Gomathi Lakshmi M and her team concentrated on practical implementations
- Each group shared key concepts on their assigned subtopic to the peer students which helped everyone learn and exchange knowledge.
- Finally, the instructor brought together all the information discussed in this activity. This not only helped students remember what they learned that day but also encouraged them to think of new ideas and answer questions about the topic more easily.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 1	2	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	2	2	1	1	1
Justification for correlation	To apply basic knowledge Underlying algorithms and protocols. Mathematical concepts are crucial in designing efficient and reliable DNS systems	To identify and analyze complex issues related to domain resolution, server configurations and security.	To design DNS should be adept at designing solutions for complex problems.	To work as an individual, and as a member	To communicate effectively on complex engineering activities	To design the FA, students will be able to develop various software components

Figure 1 illustrates a scenario in which a student is actively engaged, collaborates effectively, and attains a deep understanding of DNS through practical implementation and peer teaching.

Images / Screenshot of the practice:



Figure 1 Students sharing their key points with peer students

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students expressed that the activity helped to identify the understanding level of the concept.
- Students informed the instructor that the activity motivates them to be attentive in the class for asking doubts further.
- Student felt good, since they learn the concepts as a group. They actively participate and this activity encouraged the students to share their knowledge with others
-

❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- The activity helped the instructor to analyze, evaluate and improve their students learning.
- Students can able to explain the concepts without any confusion.
- From this activity, the students can get more clarity in the particular topic.
- Students are actively participated in each group. It helps the students are made to work in a team and share their ideas to others and also improve their communication skills. It helps to focus attention and engage students in learning. Through this activity the students can deeply understand the topic

❖ ***Challenges faced in implementation:***

- Some of the students finding difficulties in expressing the ideas
- Motivate the students those who are not involved in the activity effectively by means of addressing the benefits of collaborative learning
- Always the toppers only represent the concepts

References:

1. <https://www.edsys.in/what-is-peer-teaching/>
2. <https://www.opencolleges.edu.au/informed/features/peer-teaching/>
3. <https://tilt.colostate.edu/TipsAndGuides/Tip/180>
4. <https://www.gdrc.org/kmgmt/c-learn/>
5. <https://www.edutopia.org/topic/collaborative-learning>